

Order Escape and the Hidden Arrow of Time in Kinetic Limits

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Abstract

How can an irreversible kinetic arrow emerge from a reversible many-body or Hamiltonian flow? We propose a precise answer that separates the direction of observation from the direction of time. For any nested standard-Borel observations, we prove an extended-value chain rule: the information revealed by refining the observation is a conditional Kullback–Leibler (KL) divergence and is therefore nonnegative. Applied to an exchangeable hard-sphere state, this rule produces a nonnegative mutual-information order spectrum whose successive increments are conditional mutual informations. Under the finite-time factorization input of Deng, Hani, and Ma, together with explicitly stated entropy and canonical-transfer hypotheses, every fixed bulk order becomes asymptotically chaotic while the order-summed information per particle converges to the integrated Boltzmann dissipation. Thus the arrow is forced into orders that diverge with the particle number; no claim that these orders are proportional to the particle number is made. An exact contact-trace identity identifies the boundary channel through which this information is generated, and a change-of-measure identity connects it to a deliberately chaotic reverse preparation. As a second, solvable calibration, a single-mode wave-kinetic Fokker–Planck equation reduces after variance normalization to an Ornstein–Uhlenbeck semigroup. Its shape information decays at a Fisher-information rate, while its multiscale conditional innovations obey the same observation-chain rule. A Duhamel estimate places the hard-sphere trace error and finite-box PDF error in a common residual architecture. The resulting order–history–dissipation (OHD) framework is rigorous at the stated effective-equation level and falsifiable, but it does not by itself prove a finite-box NLS limit, a contact-trace theorem, or a nonequilibrium fluctuation theorem.

Keywords. arrow of time; Boltzmann–Grad limit; relative entropy; conditional KL increments; collision histories; wave kinetics; Ornstein–Uhlenbeck; propagation of chaos.

1 Introduction

The microscopic equations of a gas or a weakly nonlinear wave field are reversible, yet their kinetic descriptions possess a preferred temporal direction. For hard spheres, the Boltzmann equation has an increasing entropy for generic data even though the underlying collision map is an involution [3, 15]. For weakly nonlinear dispersive waves, oscillatory averaging and a hierarchy of observables produce a Markovian wave-kinetic or probability-density closure. The apparent contradiction is often summarized by saying that “molecular chaos” or “random phases” have been assumed. That phrase hides two distinct operations: discarding information carried by high-order correlations, and replacing a reversible history by an effective transition law.

The short-time benchmark is Lanford’s theorem, while recent long-time derivations of the Boltzmann equation from hard-sphere dynamics control finitely many bulk marginals on every prescribed finite interval on which the limiting solution is regular [10, 8]. Wave-kinetic derivations similarly control selected observables and higher-order statistics [5, 6, 7]. These results do not, by themselves, control an unbounded entropy functional, a collision-boundary trace, or a

path-space central limit. The present paper isolates the missing information rather than silently promoting a finite-order estimate to a full entropy theorem.

The central object is an *observation chain*. Let $Y_1 = \pi_1(Y_2) = \dots$ be increasingly detailed observations of the same random object, and compare a forward law P with a chosen reference law Q (which may be a reverse or an effective kinetic law). The KL divergence at level m is not an arbitrary sequence: its increment is an exact conditional KL. This observation fact becomes a nontrivial statement about kinetic limits when the levels are the s -particle marginals of a hard-sphere gas. It complements the classical entropy-chaos and correlation literature [9, 12]. We obtain a positive order spectrum

$$\delta_{s,\varepsilon}(t) := \mathbb{E}_{p_{s-1,\varepsilon}(t)} D_{\text{KL}}(p_{s,\varepsilon}(t, dz_s \mid z_{1:s-1}) \parallel p_{1,\varepsilon}(t, dz_s)), \quad (1)$$

whose sum is the total correlation $D_{\text{KL}}(W_{N_\varepsilon} \parallel p_{1,\varepsilon}^{\otimes N_\varepsilon})$. The main theorem below shows, conditionally and with all bridges exposed, that $\delta_{s,\varepsilon}(t) \rightarrow 0$ for every fixed s , whereas the full sum divided by N_ε tends to $\int_0^t \mathcal{D}_B[f(r)] dr$. The result is an order-escape statement: it locates the information outside every fixed kinetic order, without asserting a particular scaling for the order at which it concentrates.

The wave-kinetic side supplies a complementary calibration. For an effective two-dimensional single-mode PDF/OU closure inspired by wave-kinetic reductions (not derived here from the cited DHM theorems), the KL distance to the Gaussian with matching variance satisfies an exact Fisher-information dissipation identity. For independent OU coordinates, the joint KL splits into total correlation and marginal shape information, and each newly exposed conditional innovation is nonnegative. The signs in the two examples are intentionally different: a closed reversible gas transfers resolved one-particle information into hidden correlations, while the effective OU equation dissipates shape information after the microscopic degrees of freedom have been projected out. Confusing these two time directions would recreate the paradox we seek to resolve.

Our contributions are therefore fourfold:

- (i) an observation-chain theorem and an exchangeable mutual-information order spectrum whose successive increments are conditional innovations;
- (ii) a conditional hard-sphere order-escape theorem, including its exact information balance and the precise entropy/trace assumptions needed to connect it to the Boltzmann H theorem;
- (iii) a common residual formulation for the collision contact trace and for finite-box PDF closures, plus a path-reversal identity for a specified chaotic reverse reference;
- (iv) an analytically solvable OU/WKE benchmark and a falsifiable protocol that distinguishes bulk chaos, contact memory, and effective dissipation.

The manuscript is deliberately stratified. Results labelled theorem or proposition are proved under their displayed hypotheses. Statements about a finite-volume NLS, a nonequilibrium Boltzmann–Langevin limit, or an endpoint large-deviation rate are explicitly marked as open targets. No acceptance, priority, or authorship is implied by this synthesis.

2 Observation order is not dynamical time

Let $(\Omega, \mathcal{F})$ be a standard Borel probability space and let $P \ll Q$. Let $Y_m : \Omega \rightarrow E_m$ be standard-Borel observations satisfying $Y_m = \pi_m(Y_{m+1})$. Put $\mathcal{G}_m = \sigma(Y_m)$ and write $P_m = (Y_m)_\# P$, $Q_m = (Y_m)_\# Q$. Define

$$A_m := D_{\text{KL}}((Y_m)_\# P \parallel (Y_m)_\# Q). \quad (2)$$

Table 1: Claim ledger used throughout the paper. “Exact” means a finite-system identity; “conditional” means that the listed bridge hypotheses are additional to the kinetic-limit theorem; “target” is not claimed.

Layer	Quantity	Status here	Missing input for a stronger claim
Observation order	conditional KL increment	Exact	none beyond standard-Borel disintegration
Hard-sphere endpoint	$\sum_s \delta_{s,\varepsilon} = \mathcal{I}_{N_\varepsilon}$	Exact	finite entropy/absolute continuity
Boltzmann–Grad entropy	$\mathcal{I}_N/N \rightarrow \int \mathcal{D}_B$	Conditional	canonical transfer and uniform integrability
Collision boundary	exact current split; $R_{\text{shift}} = O_R(\varepsilon)$	exact balance; conditional shift bound	a trace convergence estimate
OU/PDF closure	Fisher dissipation	Exact for the effective PDE	finite-box residual $\rightarrow 0$
Fluctuations and echo	covariance/LDP	Target	a path-level kinetic theorem

Theorem 2.1 (Order-resolved KL chain rule). *Let $P_{m+1|m}$ and $Q_{m+1|m}$ be regular conditional kernels. Then, as an equality in $[0, \infty]$,*

$$A_{m+1} = A_m + \mathbb{E}_{P_m} \left[D_{\text{KL}}(P_{m+1|m}(\cdot | Y_m) \| Q_{m+1|m}(\cdot | Y_m)) \right]. \quad (3)$$

If $\mathcal{G}_m \uparrow \mathcal{G}_\infty$, then $A_m \uparrow D_{\text{KL}}(P|_{\mathcal{G}_\infty} \| Q|_{\mathcal{G}_\infty})$.

Proof. Write $L = dP/dQ$ and $L_m = \mathbb{E}_Q[L | \mathcal{G}_m]$. Disintegration of $L_{m+1} \log L_{m+1}$ over $\mathcal{G}_m$ gives (3); the conditional term is nonnegative by Gibbs’ inequality. The martingale convergence theorem gives $L_m \rightarrow L_\infty$ almost surely. Conditional Jensen and Fatou’s lemma, using the lower bound of $x \log x$, identify the limit of the relative entropies. $\square$

Remark 2.2. The monotonicity in Theorem 2.1 is in observation order, not in physical time. A time-reversed reference must be specified before A_m can be called a path irreversibility. In particular, comparing unrelated moments or replacing one observable by another does not produce a monotone hierarchy.

2.1 Exchangeable endpoint spectrum

Let $P_{N,s}$ be the normalized s -particle marginal of a symmetric probability density W_N on a hard-core domain, extended by zero to the full product space, and let $p_N = P_{N,1}$. Define $K_{N,1} = 0$ and, without subtracting possibly infinite quantities,

$$\delta_{N,s} := \mathbb{E}_{P_{N,s-1}} D_{\text{KL}}(P_{N,s}(dz_s | z_{1:s-1}) \| p_N(dz_s)), \quad s \geq 2, \quad (4)$$

and set $K_{N,s} := \sum_{j=2}^s \delta_{N,j}$. Whenever the KL is finite, $K_{N,s} = D_{\text{KL}}(P_{N,s} \| p_N^{\otimes s})$ by Theorem 2.1. At $s = N$, $K_{N,N}$ is the total correlation. All differential entropies below use one fixed dimensionless phase-space cell as reference, so that comparisons of h_s across particle orders are well-defined.

$$\mathcal{I}_N := D_{\text{KL}}(W_N \| p_N^{\otimes N}) = -h_N + Nh_1, \quad h_s := - \int P_{N,s} \log P_{N,s}. \quad (5)$$

Exchangeability gives a useful shape constraint. Since $\delta_{N,s} = I(Z_s; Z_{1:s-1}) = I(Z_1; Z_{2:s})$, the chain rule gives the extended-value identity

$$\delta_{N,s+1} = \delta_{N,s} + I(Z_1; Z_{s+1} | Z_{2:s}),$$

and hence nonnegativity of conditional mutual information implies

$$0 \leq \delta_{N,2} \leq \delta_{N,3} \leq \cdots \leq \delta_{N,N}. \quad (6)$$

Thus the increments are nondecreasing (and $K_{N,s}$ is discrete convex whenever the displayed quantities are finite). This positivity is different from the signed entropy cumulants κ_s^{ent} . For finite entropies, define

$$\kappa_s^{\text{ent}} := \sum_{j=1}^s (-1)^{s-j} \binom{s}{j} h_j, \quad (7)$$

and then

$$K_{N,s} = - \sum_{r=2}^s \binom{s}{r} \kappa_r^{\text{ent}}, \quad \delta_{N,s} = - \sum_{r=2}^s \binom{s-1}{r-1} \kappa_r^{\text{ent}}, \quad (8)$$

when the displayed entropies are finite. Equation (4) is the preferred definition because every order increment is manifestly nonnegative.

2.2 Path observations

The same theorem applies to a forward path law P^{F} and a chosen reverse or effective law Q^{R} . For example, one may take Y_0 to be a one-particle endpoint, then add fixed-order endpoints, a contact trace, and finally a collision-history record. The chain rule gives

$$A_{m+1}^{\text{path}} = A_m^{\text{path}} + \mathbb{E}_{P_m^{\text{path}}} D_{\text{KL}} \left(P^{\text{F}}(dY_{m+1} | Y_m) \parallel Q^{\text{R}}(dY_{m+1} | Y_m) \right) \geq A_m^{\text{path}}, \quad (9)$$

where the equality is in the extended nonnegative reals (and can be read as an ordinary difference when $A_m^{\text{path}} < \infty$). This is an exact information budget. A finite-order endpoint estimate does not imply that a finite-order path KL is small; such a transfer requires a separate path-level uniform-integrability or trace theorem.

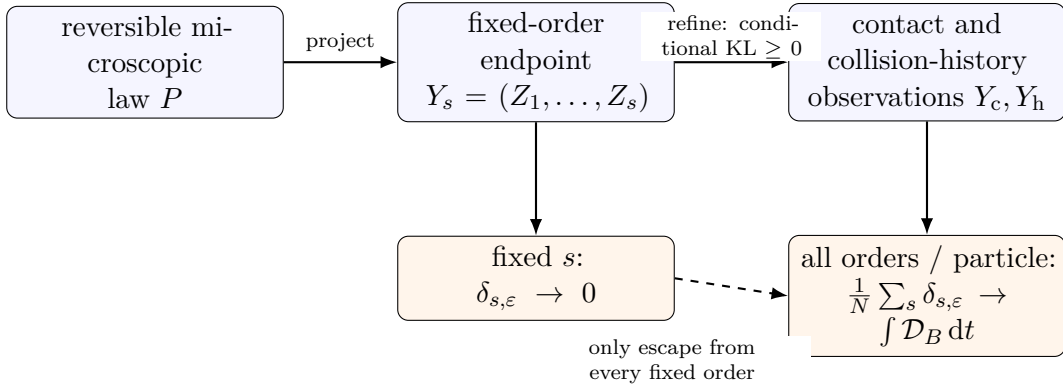


Figure 1: The two axes of the information ledger. Refining an observation adds a nonnegative conditional KL; evolving in time has model-dependent signs. In the hard-sphere limit, fixed bulk orders lose their information while the order-summed endpoint information per particle approaches the Boltzmann dissipation. The dashed implication is deliberately only an escape-to-diverging-orders statement.

3 Hard spheres: an order-escape theorem

3.1 Microscopic setting and kinetic input

Consider N_ε identical hard spheres of diameter ε in $\mathbf{X}^d \times \mathbb{R}^d$, $d \geq 2$, where $\mathbf{X}^d$ is either $\mathbb{R}^d$ or a periodic torus (the boundary terms below are interpreted accordingly). For every $n \geq 0$ define

the hard-core domain

$$\mathcal{D}_n^\varepsilon := \{z_n : |x_i - x_j|_{\mathbf{X}^d} \geq \varepsilon, \ 1 \leq i < j \leq n\}, \quad z_j = (x_j, v_j), \quad (10)$$

and use $\mathcal{D}_{N_\varepsilon}^\varepsilon$ for the N_ε -particle phase space; in particular, $\mathcal{D}_s^\varepsilon$ in the marginal estimates is the case $n = s$ of this definition. At a binary contact $x_j = x_i + \varepsilon\omega$, with $u = v_i - v_j$ and $u \cdot \omega > 0$ on the incoming set, the collision map is

$$v'_i = v_i - (u \cdot \omega)\omega, \quad v'_j = v_j + (u \cdot \omega)\omega. \quad (11)$$

It is an involution preserving momentum, kinetic energy, and phase-space volume. The flow $H_{N_\varepsilon}^\varepsilon(t)$ obeys, away from the usual null set of grazing or multiple collisions,

$$H_{N_\varepsilon}^\varepsilon(-t) = \Theta H_{N_\varepsilon}^\varepsilon(t) \Theta, \quad \Theta(x, v) = (x, -v). \quad (12)$$

The Boltzmann–Grad scaling is

$$N_\varepsilon \varepsilon^{d-1} \longrightarrow \alpha \in (0, \infty). \quad (13)$$

For notational simplicity set $\alpha = 1$ in the displayed kinetic equation; the general- α convention is stated after the order-escape theorem. The limiting probability density $f_t = f(t, x, v)$ solves

$$(\partial_t + v \cdot \nabla_x) f = Q(f, f), \quad Q(f, f) = \int_{\mathbb{R}^d \times \mathbb{S}^{d-1}} [(v - v_1) \cdot \omega]_+ (f' f'_1 - f f_1) dv_1 d\omega. \quad (14)$$

Here $f_1 = f(t, x, v_1)$, $f' = f(t, x, v')$, and $f'_1 = f(t, x, v'_1)$, with the collision velocities given by (11); all collision integrals below also include dx . We use $\int f_t dx dv = 1$ and impose periodic spatial boundary conditions or sufficient decay at infinity, so that the free-transport term does not contribute to the entropy balance. For the fixed- N argument we formulate the needed finite-time factorization input directly as the following canonicalized estimate, for each fixed T in the regular lifespan of f and each fixed s (or uniformly for $s \leq |\log \varepsilon|$ in the native normalization),

$$\sup_{0 \leq t \leq T} \|p_{s,\varepsilon}(t) - f_t^{\otimes s} \mathbf{1}_{\mathcal{D}_s^\varepsilon}\|_{L^1} \leq C_{T,s} \varepsilon^\theta. \quad (15)$$

Here $p_{s,\varepsilon}$ denotes a normalized canonical marginal. This is a canonicalized form of the finite-time factorial estimate proved by Deng, Hani, and Ma (DHM), not a claim that the DHM theorem itself supplies normalized canonical marginals. The transfer from their native grand-canonical correlations to this $p_{s,\varepsilon}$, including the entropy bridge, is isolated below as an independent assumption. For $\mathbf{X}^d = \mathbb{R}^d$, (15) is the DHM input being invoked; the periodic-torus version is an additional analogue assumption (and is not attributed to that theorem). For the fixed- N_ε formulas below we identify $P_{N_\varepsilon, s}^\varepsilon = p_{s,\varepsilon}$ and write $p_{N_\varepsilon} = p_{1,\varepsilon}$.

Assumption 3.1 (Entropy bridge). On $[0, T]$, for every fixed s , $p_{s,\varepsilon} \ll p_{1,\varepsilon}^{\otimes s}$ and the corresponding global KL divergences are finite. (Equivalently, one may define them as monotone limits of truncated KLs and assume that the limits are finite.) The L^1 convergence in (15) is accompanied by uniform integrability of the absolute values $|p_{s,\varepsilon} \log p_{s,\varepsilon}|$ and $|p_{s,\varepsilon} \log(p_{1,\varepsilon}^{\otimes s})|$, and by $p_{1,\varepsilon} \rightarrow f_t$ in the corresponding one-particle entropy and cross-entropy topologies. On every spatial/velocity truncation set K_R used below, the hard-core product defect satisfies

$$\int_{K_R^s} f_t^{\otimes s} (1 - \mathbf{1}_{\mathcal{D}_s^\varepsilon}) dz_s = O_R(s^2 \varepsilon^d) \quad (16)$$

uniformly in t , and the complementary tails vanish uniformly as $R \rightarrow \infty$. Here and below K_R denotes a fixed compact truncation in the variables being integrated (including x when $\mathbf{X}^d = \mathbb{R}^d$; on a torus only the velocity cutoff is nontrivial).

Assumption 3.2 (Canonical transfer and initial entropy chaos). The canonical normalization and the grand-canonical-to-canonical conversion preserve (15) and the entropy bridge for each fixed order, and the initial hard-core product preparation obeys $\mathcal{I}_{N_\varepsilon}(0)/N_\varepsilon \rightarrow 0$. A sufficient low-packing estimate is $|\log Z_{N_\varepsilon, 0}| \leq C N_\varepsilon^2 \varepsilon^d$ together with the corresponding one-particle tail bounds. Here the preparation means the normalized conditioned product law $W_{N_\varepsilon}^\varepsilon(0) \propto \mathbf{1}_{\mathcal{D}_{N_\varepsilon}^\varepsilon} \prod_{j=1}^{N_\varepsilon} f_0(z_j)$, with the same spatial and velocity convention as above.

3.2 Native grand-canonical alternative

The factorial normalization used in the kinetic expansion is naturally a law on all particle-number sectors. Let $\mathfrak{X}^\varepsilon = \bigsqcup_{n \geq 0} (X^d \times \mathbb{R}^d)^n$ and use $\nu^\varepsilon|_n = dz_n/n!$ as reference measure; hard-core densities are extended by zero outside $\mathcal{D}_n^\varepsilon$. If P_t^ε has sector densities W_n^ε that are symmetric within each sector, define the density convention by $\sum_{n \geq 0} (1/n!) \int W_n^\varepsilon(t, z_n) dz_n = 1$ and

$$\bar{N}_\varepsilon := \sum_{n \geq 1} \frac{1}{n!} \int n W_n^\varepsilon(t, z_n) dz_n, \quad m_\varepsilon(t, z) := \sum_{n \geq 1} \frac{1}{(n-1)!} \int W_n^\varepsilon(t, z, z_{2:n}) dz_{2:n}, \quad (17)$$

$$\rho_\varepsilon(t, z) := m_\varepsilon(t, z)/\bar{N}_\varepsilon, \quad h_{\text{gc}}[P_t^\varepsilon] := - \sum_{n \geq 0} \frac{1}{n!} \int W_n^\varepsilon \log W_n^\varepsilon dz_n. \quad (18)$$

For a probability density ρ and mean $\lambda > 0$, the Poisson reference has sector density

$$\Pi_{\lambda, \rho, n}(z_n) := e^{-\lambda} \lambda^n \prod_{j=1}^n \rho(z_j). \quad (19)$$

Write $\Pi_{\lambda, \rho} := (\Pi_{\lambda, \rho, n})_{n \geq 0}$ and $h[\rho] := - \int \rho(z) \log \rho(z) dz$. For sector densities $U = (U_n)$ and $V = (V_n)$, write

$$D_{\text{KL}}^{\nu^\varepsilon}(U \| V) := \sum_{n \geq 0} \frac{1}{n!} \int U_n(z_n) \log \frac{U_n(z_n)}{V_n(z_n)} dz_n, \quad (20)$$

with value $+\infty$ when absolute continuity fails.

Proposition 3.3 (Native Poisson information balance). *Assume $0 < \bar{N}_\varepsilon < \infty$, that P_t^ε has the normalized sector densities W_n^ε with respect to ν^ε in (18), and that each sector is transported by a volume-preserving hard-sphere flow (so that particle-number sectors do not mix). Assume also that the relevant entropies are finite and that P_t^ε is absolutely continuous with respect to the sectorwise Poisson law $\Pi_{\bar{N}_\varepsilon, \rho_\varepsilon(t)}$. Then*

$$\mathcal{I}_{\text{gc}, \varepsilon}(t) := D_{\text{KL}}^{\nu^\varepsilon}(P_t^\varepsilon \| \Pi_{\bar{N}_\varepsilon, \rho_\varepsilon(t)}) = -h_{\text{gc}}[P_t^\varepsilon] + \bar{N}_\varepsilon(1 - \log \bar{N}_\varepsilon) + \bar{N}_\varepsilon h[\rho_\varepsilon(t)], \quad (21)$$

and Liouville transport gives

$$\frac{\mathcal{I}_{\text{gc}, \varepsilon}(t) - \mathcal{I}_{\text{gc}, \varepsilon}(0)}{\bar{N}_\varepsilon} = h[\rho_\varepsilon(t)] - h[\rho_\varepsilon(0)]. \quad (22)$$

More generally, for any Poisson reference with mean $\lambda > 0$ and one-particle density σ ,

$$D_{\text{KL}}^{\nu^\varepsilon}(P_t^\varepsilon \| \Pi_{\lambda, \sigma}) - \mathcal{I}_{\text{gc}, \varepsilon}(t) = \lambda - \bar{N}_\varepsilon - \bar{N}_\varepsilon \log \frac{\lambda}{\bar{N}_\varepsilon} + \bar{N}_\varepsilon D_{\text{KL}}(\rho_\varepsilon(t) \| \sigma) \geq 0. \quad (23)$$

For the hard-core product preparation with $\lambda_\varepsilon = \varepsilon^{-(d-1)}$, let the initial law P_0^ε have sector densities

$$W_{0,n}^\varepsilon(z_n) := Z_{\text{gc}, \varepsilon}^{-1} \lambda_\varepsilon^n \mathbf{1}_{\mathcal{D}_n^\varepsilon}(z_n) \prod_{j=1}^n f_0(z_j), \quad Z_{\text{gc}, \varepsilon} := \sum_{n \geq 0} \frac{\lambda_\varepsilon^n}{n!} \int_{\mathcal{D}_n^\varepsilon} \prod_{j=1}^n f_0(z_j) dz_n. \quad (24)$$

Then $0 \leq \mathcal{I}_{\text{gc}, \varepsilon}(0) \leq \lambda_\varepsilon - \log Z_{\text{gc}, \varepsilon}$. Consequently, if $\bar{N}_\varepsilon \sim \lambda_\varepsilon$, $\lambda_\varepsilon - \log Z_{\text{gc}, \varepsilon} = o(\lambda_\varepsilon)$, and $h[\rho_\varepsilon(t)] \rightarrow s_B[f_t]$ uniformly on $[0, T]$ (including the initial time $t = 0$), then

$$\frac{\mathcal{I}_{\text{gc}, \varepsilon}(t)}{\bar{N}_\varepsilon} \longrightarrow s_B[f_t] - s_B[f_0] = \int_0^t \mathcal{D}_B[f_r] dr. \quad (25)$$

Proof. Expanding the logarithm in (21) and using the sector probabilities gives the three displayed terms; volume preservation keeps h_{gc} and $\bar{N}_\varepsilon$ constant. The initial bound follows by comparing with the unconstrained Poisson law and minimizing over its mean and one-particle density. The final limit follows from (22), Boltzmann's H theorem, and the stated acceptance estimate. $\square$

This route avoids a de-Poissonization step, but its acceptance and entropy-tail conditions are independent hypotheses; they are not consequences of the bare factorial L^1 estimate. The fixed- N theorem below is therefore stated separately rather than silently identifying the two ensembles.

3.3 Collision trees and connected cumulants

The DHM history expansion separates tree-like collision histories, which feed the limiting Boltzmann operator, from non-tree molecules and recollisions. In the information ledger the former are resolved by the kinetic transition kernel, whereas the latter are natural carriers of the hidden history budget. If $c_{N,s}^\varepsilon$ denotes the ordinary connected density cumulant, defined by the partition-lattice relation

$$p_{s,\varepsilon}(z_s) = \sum_{\pi \in \Pi_s} \prod_{B \in \pi} c_{N,|B|}^\varepsilon(z_B), \quad c_{N,1}^\varepsilon = p_{1,\varepsilon}, \quad (26)$$

Under the local boundedness/tightness hypotheses needed for the finite-order cumulant inversion, the canonical transfer and excluded-volume conversion give, for every fixed s and fixed truncation K_R , the representative bound

$$\|c_{N,s}^\varepsilon\|_{L^1(K_R^s)} \leq C_{T,R,s}(\varepsilon^\theta + s^2 \varepsilon^d). \quad (27)$$

This is a finite-order bulk statement (and is itself a conditional local estimate); it does not control the contact trace, the non-tree history law, or the order-summed KL.

For bounded one-particle observables with an explicit compact truncation (or with a separately verified uniform tail bound), one can still obtain a finite-resolution check. Let $\zeta_{N,t}(F) = N^{-1} \sum_{j=1}^N F(Z_j(t))$, where F is bounded and supported in a fixed one-particle truncation K_R , and $c_{N,2}^\varepsilon = p_{2,\varepsilon} - p_{1,\varepsilon}^{\otimes 2}$. Symmetry gives the exact variance decomposition

$$\text{Var } \zeta_{N,t}(F) = \frac{1}{N} \left[\int F^2 p_{1,\varepsilon} - \left(\int F p_{1,\varepsilon} \right)^2 \right] + \frac{N-1}{N} \iint F(z) F(z') c_{N,2}^\varepsilon(z, z') dz dz'. \quad (28)$$

Together with the fixed-order bound above and the corresponding tail estimate, this controls the truncated bounded-observable means and supplies a concrete fluctuation benchmark. Without that tail estimate it is only a finite-resolution identity. In either case it does not imply a central-limit theorem: at the kinetic scaling the conservative L^1 error in the pair term can exceed the $N^{-1/2}$ fluctuation scale, so a Boltzmann–Langevin limit remains an open target.

3.4 The order-escape theorem

Theorem 3.4 (Order escape and the hidden-arrow balance). *Let $W_{N_\varepsilon}^\varepsilon(t)$ be a symmetric hard-sphere law transported by the volume-preserving flow, with $W_{N_\varepsilon}^\varepsilon \ll p_{N_\varepsilon}^{\otimes N_\varepsilon}$. For the finite-entropy statements below assume finite total correlation (the order-sum identity in part (b) has an extended-value interpretation if this is dropped). Assume that its initial law is the hard-core product preparation covered by Assumption 3.2, with finite initial total correlation and $\mathcal{I}_{N_\varepsilon}^\varepsilon(0)/N_\varepsilon \rightarrow 0$. Define $\delta_{N_\varepsilon,s}^\varepsilon(t)$ and $K_{N_\varepsilon,s}^\varepsilon(t)$ by (4). Under Assumptions 3.1 and 3.2, and the canonicalized finite-time factorization input (15), the following hold.*

(a) For every fixed s , uniformly on $[0, T]$,

$$K_{N_\varepsilon, s}^\varepsilon(t) \longrightarrow 0, \quad \delta_{N_\varepsilon, s}^\varepsilon(t) \longrightarrow 0. \quad (29)$$

(b) The exact finite- N identity is

$$\mathcal{I}_{N_\varepsilon}^\varepsilon(t) = K_{N_\varepsilon, N_\varepsilon}^\varepsilon(t) = \sum_{s=2}^{N_\varepsilon} \delta_{N_\varepsilon, s}^\varepsilon(t), \quad \frac{\mathcal{I}_{N_\varepsilon}^\varepsilon(t) - \mathcal{I}_{N_\varepsilon}^\varepsilon(0)}{N_\varepsilon} = h_{1, \varepsilon}(t) - h_{1, \varepsilon}(0). \quad (30)$$

The decomposition into the order sum is valid in the extended nonnegative reals; the displayed difference of entropies is asserted when the relevant finite entropies are finite.

(c) If, in addition, the one-particle entropy converges to the Boltzmann entropy $s_B[f_t] = -\int f_t \log f_t \, dx \, dv$ and the normalized finite entropies in (30) are well-defined, then

$$\frac{\mathcal{I}_{N_\varepsilon}^\varepsilon(t)}{N_\varepsilon} \longrightarrow S_B(t) := s_B[f_t] - s_B[f_0] = \int_0^t \mathcal{D}_B[f_r] \, dr, \quad (31)$$

where

$$\mathcal{D}_B[f] := \frac{1}{4} \int B(f' f'_1 - f f_1) \log \frac{f' f'_1}{f f_1} \, dx \, dv \, dv_1 \, d\omega \geq 0, \quad B = [(v - v_1) \cdot \omega]_+. \quad (32)$$

The displayed normalization uses $\alpha = 1$. For a general scaling limit $N_\varepsilon \varepsilon^{d-1} \rightarrow \alpha$, let f^α solve $(\partial_t + v \cdot \nabla_x) f^\alpha = \alpha Q(f^\alpha, f^\alpha)$ and define

$$S_B^\alpha(t) := s_B[f_t^\alpha] - s_B[f_0^\alpha] = \int_0^t \alpha \mathcal{D}_B[f_r^\alpha] \, dr. \quad (33)$$

Then the corresponding limits in (31) and below use S_B^α (the formulas displayed above are the case $\alpha = 1$). In the remainder of the displayed theorem we keep the unit-rate convention, so the positivity condition and (34) use S_B . Consequently, for every fixed integer S and every t with $S_B(t) > 0$,

$$\frac{1}{N_\varepsilon} \sum_{s=S+1}^{N_\varepsilon} \delta_{N_\varepsilon, s}^\varepsilon(t) \longrightarrow S_B(t). \quad (34)$$

For $\alpha \neq 1$, replace S_B by S_B^α in this tail statement.

In particular, the information cannot remain in any fixed kinetic order. The theorem does not determine whether the carrying order is $o(N_\varepsilon)$, proportional to N_ε , or concentrated in a sparse set of diverging orders.

Proof. Theorem 2.1 applied to the endpoint projections gives the nonnegative decomposition in (4) and the first equality in (30); the second follows from Liouville invariance of h_{N_ε} . Under Assumption 3.1, truncating $u \log u$ and the cross entropy on a compact velocity set, using (15) and (16), yields

$$D_{\text{KL}}(p_{s, \varepsilon}(t) \| p_{1, \varepsilon}(t)^{\otimes s}) \rightarrow 0$$

for every fixed s , uniformly in t . This proves (a). The same entropy bridge gives $h_{1, \varepsilon}(t) \rightarrow s_B[f_t]$; inserting it into (30) and using the Boltzmann H theorem gives (c). Finally, the finite sum $K_{N_\varepsilon, S} = \sum_{s=2}^S \delta_{N_\varepsilon, s}$ is $o(N_\varepsilon)$ by (a), so subtracting it from $K_{N_\varepsilon, N_\varepsilon} = \mathcal{I}_{N_\varepsilon}$ proves (34). $\square$

Remark 3.5. The theorem is intentionally conditional. The bare L^1 estimate in (15) does not control the unbounded logarithm in a KL divergence. Nor does a grand-canonical factorial estimate automatically supply the canonical transfer in Assumption 3.2. These are analytic tasks, not cosmetic qualifications.

Proposition 3.6 (Reference splitting). *For any one-particle reference density r_t for which the terms are finite,*

$$D_{\text{KL}}(P_{N,s}(t) \| r_t^{\otimes s}) = K_{N,s}(t) + s D_{\text{KL}}(p_N(t) \| r_t). \quad (35)$$

Thus the product reference p_N isolates pure correlation information, whereas the Boltzmann reference f_t adds a one-particle closure or shape defect. This is the exact analogue of the total-correlation plus marginal-shape decomposition used below for OU modes.

3.5 The contact channel and the temporal rate

Bulk L^1 convergence does not determine a two-particle density on the collision boundary. Let $\alpha_N = (N-1)\varepsilon^{d-1}$ and denote the incoming boundary trace, on $u \cdot \omega > 0$, by

$$p_{N,2}^{\varepsilon,-}(t, x, v, v_1, \omega) := \text{Tr}_{x_2=x+\varepsilon\omega}^- p_{N,2}^{\varepsilon}(t, x, v, x_2, v_1). \quad (36)$$

Thus the trace is taken from the admissible side of the hard-core boundary, with the incoming velocity set fixed as above, and

$$\gamma_{N,\varepsilon}^-(t, x, v, v_1, \omega) := \alpha_N p_{N,2}^{\varepsilon,-}(t, x, v, v_1, \omega). \quad (37)$$

Put

$$A_{N,\varepsilon} := p_N(x, v) p_N(x + \varepsilon\omega, v_1), \quad (38)$$

$$A'_{N,\varepsilon} := p_N(x, v') p_N(x + \varepsilon\omega, v'_1), \quad (39)$$

$$\ell_{N,\varepsilon} := \log(A_{N,\varepsilon}/A'_{N,\varepsilon}). \quad (40)$$

Proposition 3.7 (Exact contact-trace entropy identity). *Assume periodic space (or sufficient decay), positive C^1 one-particle marginals, incoming and outgoing traces, and integrable boundary entropy fluxes $Bp_{N,2}^{\varepsilon,\pm}(1 + |\log p_N|)$ together with $B\gamma_{N,\varepsilon}^-|\ell_{N,\varepsilon}| \in L^1$. (An equivalent renormalized trace formulation may be used.) Then*

$$\frac{d}{dt} h_{1,\varepsilon}(t) = \frac{1}{2} \int B\gamma_{N,\varepsilon}^- \ell_{N,\varepsilon} \, dx \, dv \, dv_1 \, d\omega \quad (41)$$

$$= \alpha_N J_{N,\varepsilon}[p_N] + R_{\text{contact}}(t), \quad (42)$$

where

$$J_{N,\varepsilon}[p] := \frac{1}{2} \int B A_{N,\varepsilon} \ell_{N,\varepsilon} \, dx \, dv \, dv_1 \, d\omega, \quad (43)$$

$$R_{\text{contact}}(t) := \frac{1}{2} \int B(\gamma_{N,\varepsilon}^- - \alpha_N A_{N,\varepsilon}) \ell_{N,\varepsilon} \, dx \, dv \, dv_1 \, d\omega. \quad (44)$$

If, on every fixed cutoff K_R , all pre- and post-collisional factors entering $A_{N,\varepsilon}$ and $A'_{N,\varepsilon}$ are bounded between constants $0 < c_R \leq C_R < \infty$, their spatial logarithmic gradients are uniformly bounded, and the corresponding B -weighted integrals are uniformly finite (with a separately controlled tail outside K_R), then

$$J_{N,\varepsilon}[p_N] = \mathcal{D}_B[p_N] + R_{\text{shift}}(t), \quad R_{\text{shift}} = O_R(\varepsilon) \quad (45)$$

on the cutoff, with a tail remainder after the cutoff is removed. The shifted product term need not be nonnegative at finite diameter; nonnegativity belongs to the local Boltzmann dissipation.

Proof. The first BBGKY equation has boundary flux $(v_1 - v) \cdot \omega p_{N,2}$ at $x_2 = x + \varepsilon\omega$. The specular collision map changes the outgoing trace into the incoming trace and reverses the sign of $u \cdot \omega$. Multiplication by $-(1 + \log p_N)$, integration, and symmetrization give the first line of (42). Adding and subtracting the product trace gives (44). A first-order spatial Taylor expansion of $p_N(x + \varepsilon\omega, v_1)$ yields (45); the local pre/post-collisional change of variables gives (32). $\square$

For clarity, $\mathcal{D}_B[p]$ in this finite- N formula denotes the local Boltzmann functional obtained from (32) by replacing f with p :

$$\mathcal{D}_B[p] := \frac{1}{4} \int B(p'p'_1 - pp_1) \log \frac{p'p'_1}{pp_1} dx dv dv_1 d\omega. \quad (46)$$

Combining Proposition 3.7 with (30) gives the finite- N information-current identity

$$\frac{d}{dt} \frac{\mathcal{I}_N^\varepsilon(t)}{N} = \alpha_N \mathcal{D}_B[p_N(t)] + R_{\text{contact}}(t) + \alpha_N R_{\text{shift}}(t). \quad (47)$$

The proposed trace estimate

$$\sup_{t \leq T} \int_{K_R^x} \int_{|v|+|v_1| \leq R} B|\gamma_{N,\varepsilon}^- - \alpha_N A_{N,\varepsilon}| |\ell_{N,\varepsilon}| dx dv dv_1 d\omega \leq C_{T,R} \varepsilon^\theta \quad (48)$$

would close the particle-side residual budget. Here K_R^x is a spatial compact (and is the whole torus in the periodic case). An unweighted trace estimate would also suffice if p_N and its post-collisional values have uniform upper and positive lower bounds on the cutoff, with a separately controlled tail. Neither statement is a consequence of bulk factorization, and both are left as concrete theorem targets.

3.6 A specified reverse experiment

Let $\Gamma = (Z_N(t))_{0 \leq t \leq T}$, let $W_{N,0}$ be the forward initial density, and choose a normalized reverse preparation $Q_{N,T}$. Define

$$\Sigma_{N,T}^Q(\Gamma) := \log \frac{W_{N,0}(Z_N(0))}{Q_{N,T}(\Theta Z_N(T))}. \quad (49)$$

Let P_F be the path law generated from $W_{N,0}$ by the forward flow, and let P_R^Q be the law generated from $Q_{N,T}$ by the reversed flow. Assume mutual absolute continuity on the transported supports and write $\Gamma^\dagger(t) = \Theta\Gamma(T-t)$; on paired paths set $\Sigma_R^Q(\Gamma^\dagger) = -\Sigma_{N,T}^Q(\Gamma)$. Volume preservation and reversibility then imply the exact integral fluctuation relation and Crooks-type relation (with p_R denoting the law under P_R^Q), in the standard change-of-measure sense [4, 13]

$$\mathbb{E}_F e^{-\Sigma_{N,T}^Q} = 1, \quad p_F(s) = e^s p_R(-s), \quad (50)$$

for the paired reverse experiment. A nonzero arrow requires a choice of $Q_{N,T}$. For the chaotic reverse preparation

$$Q_{N,T}(y_N) = Z_{N,T}^{-1} \mathbf{1}_{\mathcal{D}_N^\varepsilon}(y_N) \prod_{j=1}^N f_T(\Theta y_j), \quad (51)$$

Assume in addition the endpoint conditions $(\log Z_{N_\varepsilon,T})/N_\varepsilon \rightarrow 0$ and $D_{\text{KL}}(p_{1,N_\varepsilon}(T) \| f_T) \rightarrow 0$, together with the initial entropy-chaos and cross-entropy hypotheses stated above. Since the ratio in (49) evaluates $Q_{N,T}$ at $\Theta Z_N(T)$, the two velocity reversals in (51) cancel; equivalently, one may use the endpoint variable $z = \Theta y$. The endpoint entropy bridge then gives, conditionally,

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{N_\varepsilon} \mathbb{E}_F \Sigma_{N_\varepsilon,T}^Q = \int f_0 \log f_0 - \int f_T \log f_T = \int_0^T \mathcal{D}_B[f_t] dt. \quad (52)$$

If instead $Q_{N,T}(y) = W_N^\varepsilon(T, \Theta y)$, then $\Sigma^Q \equiv 0$. The coarse graining is therefore located in the reverse preparation, not in a violation of microscopic reversibility. Equation (52) is an endpoint statement; a path-level convergence theorem remains open.

The state and path versions can be compared before taking any limit. For the chaotic preparation (51), direct expansion gives

$$\frac{1}{N} \mathbb{E}_F \Sigma_{N,T}^Q = -\frac{h_N(0)}{N} + \frac{\log Z_{N,T}}{N} - \int p_{1,N}(T, z) \log f_T(z) dz. \quad (53)$$

With the same velocity convention as in (51),

$$\frac{1}{N} \mathbb{E}_F \Sigma_{N,T}^Q - \frac{\mathcal{I}_N(T)}{N} = \frac{\log Z_{N,T}}{N} + D_{\text{KL}}(p_{1,N}(T) \| f_T). \quad (54)$$

If the endpoint is parametrized in the reversed coordinate $\bar{z} = \Theta z$, set $\bar{p}_{1,N}(T, \bar{z}) = p_{1,N}(T, \Theta \bar{z})$. The identical bridge is then

$$\frac{1}{N} \mathbb{E}_F \Sigma_{N,T}^Q - \frac{\mathcal{I}_N(T)}{N} = \frac{\log Z_{N,T}}{N} + D_{\text{KL}}(\bar{p}_{1,N}(T) \| f_T \circ \Theta), \quad (f_T \circ \Theta)(\bar{z}) = f_T(\Theta \bar{z}). \quad (55)$$

Thus endpoint normalization and one-particle closure are the only gaps between the path arrow and the hidden state information. Under their vanishing conditions, both quantities have the same Boltzmann–Grad limit. The formula also makes clear why the actual reversed final law gives zero: it changes the reference on the right-hand side rather than changing the microscopic flow.

4 Wave/PDF calibration: the same order chain with a different time sign

4.1 Single-mode moving Gaussian

We choose the following effective calibration model, inspired by wave-kinetic PDF reductions [5, 6] but not asserted here to follow from the cited DHM theorems. Consider a positive probability density ρ_τ on $\mathbb{R}^2$ satisfying

$$\partial_\tau \rho = \frac{\sigma(\tau)}{4} \Delta \rho - \frac{\Gamma(\tau)}{2} \nabla \cdot (v \rho), \quad \sigma(\tau) \geq 0, \quad (56)$$

with sufficient decay, and let

$$n(\tau) = \int |v|^2 \rho_\tau(v) dv, \quad G_n(v) = \frac{1}{\pi n} e^{-|v|^2/n}, \quad (57)$$

so G_n is the isotropic Gaussian with the same radial second moment (and, under the zero-mean hypothesis below, the same centered second moment).

Theorem 4.1 (Moving-Gaussian dissipation and OU reduction). *Assume that $\rho_\tau > 0$ has finite second moment and finite Fisher information, with sufficient decay (uniformly on compact time intervals) to justify the differentiations and integrations by parts below. Assume also that the initial mean is zero, so that $\int v \rho_\tau(v) dv = 0$ for all τ (otherwise the reference Gaussian must be centered at the evolving mean). Let $H(\tau) = D_{\text{KL}}(\rho_\tau \| G_{n(\tau)})$ and*

$$I_{\text{rel}}(\rho \| G_n) := \int \rho |\nabla \log(\rho/G_n)|^2 dv. \quad (58)$$

Then $\dot{n} = \sigma + \Gamma n$ and

$$\dot{H}(\tau) = -\frac{\sigma(\tau)}{4} I_{\text{rel}}(\rho_\tau \| G_{n(\tau)}) \leq 0. \quad (59)$$

With the effective clock $s(\tau) = \int_0^\tau \sigma(r)/n(r) dr$ and $q_{s(\tau)}(w) = n(\tau) \rho_\tau(\sqrt{n(\tau)} w)$, the normalized density solves, in the time-change sense (classically on intervals where $\dot{s} > 0$),

$$\partial_s q = \frac{1}{4} \Delta q + \frac{1}{2} \nabla \cdot (w q), \quad G_1(w) = \pi^{-1} e^{-|w|^2}. \quad (60)$$

If the mean is zero and the cumulants exist, then $\kappa_m(W_s) = e^{-ms/2} \kappa_m(W_0)$ for every $m \geq 3$, and $H(\tau) \leq e^{-s(\tau)} H(0)$.

Proof. Integration by parts gives $\dot{n} = \sigma + \Gamma n$ and $\dot{h}(\rho) = \sigma I(\rho)/4 + \Gamma$, where $I(\rho) = \int \rho |\nabla \log \rho|^2$. Since $H = \log(\pi en) - h(\rho)$ and $I_{\text{rel}} = I - 4/n$, (59) follows from $nI \geq 4$. The change of variables to (s, w) cancels the drift and gives (60). The Mehler representation $W_s \stackrel{\text{law}}{=} e^{-s/2} W_0 + \sqrt{1 - e^{-s}} Z$, $Z \sim G_1$, proves the cumulant formula. The Gaussian logarithmic Sobolev inequality at variance n gives $I_{\text{rel}} \geq 4H/n$, hence $\dot{H} \leq -\dot{s} H$ and the exponential bound. $\square$

Proposition 4.2 (Product OU layers). *Let $q_t > 0$ be a sufficiently smooth, rapidly decaying probability density on $(\mathbb{R}^2)^r$, with all displayed KL and Fisher informations finite and with enough regularity to justify differentiation and integration by parts. Let q_t solve*

$$\partial_t q_t = \sum_{j=1}^r a_j(t) \left(\frac{1}{4} \Delta_j q_t + \frac{1}{2} \nabla_j \cdot (w_j q_t) \right), \quad a_j(t) \geq 0, \quad (61)$$

Assume that the marginals have zero mean and covariance $\text{Cov}(w_j, 0) = \frac{1}{2} I_2$ initially. This covariance condition is preserved by the OU evolution, so $\text{Cov}(w_j, t) = \frac{1}{2} I_2$ and $\mathbb{E}|w_j|^2 = 1$ for all t . Define

$$\text{TC} := D_{\text{KL}} \left(q_t \left\| \prod_j q_{j,t} \right\| \right), \quad \text{SH} := \sum_j D_{\text{KL}}(q_{j,t} \| G_1), \quad \text{JH} := D_{\text{KL}}(q_t \| G_1^{\otimes r}). \quad (62)$$

Then $\text{JH} = \text{TC} + \text{SH}$ and

$$\dot{\text{TC}} = -\frac{1}{4} \sum_j a_j (J_j - I_j) \leq 0, \quad (63)$$

$$\dot{\text{SH}} = -\frac{1}{4} \sum_j a_j (I_j - 4) \leq 0, \quad (64)$$

where $I_j = \int q_{j,t} |\nabla \log q_{j,t}|^2$ and $J_j = \int q_t |\nabla_j \log q_t|^2$. For the marginal on the first m modes, set $H_m = D_{\text{KL}}(q_t^{(m)} \| G_1^{\otimes m})$ and $H_0 = 0$ (the empty marginal). Then $B_m = H_m - H_{m-1}$ satisfies

$$B_m = \mathbb{E}_{q_t^{(m-1)}} D_{\text{KL}} \left(q_t^{(m)} (dw_m | w_{1:m-1}) \| G_1 \right) \geq 0. \quad (65)$$

Writing $J_\ell^{(m)} = \int q_t^{(m)} |\nabla_\ell \log q_t^{(m)}|^2 dw$ for the partial Fisher information of the m -mode marginal, the same calculation gives the resolved innovation balance

$$\dot{B}_m = -\frac{1}{4} \left[a_m (J_m^{(m)} - 4) + \sum_{\ell < m} a_\ell (J_\ell^{(m)} - J_\ell^{(m-1)}) \right] \leq 0. \quad (66)$$

Proof. The decomposition follows by expanding the logarithmic density ratio. Entropy differentiation and conditional Jensen give (64); the two-dimensional Cauchy–Schwarz inequality gives $I_j \geq 4$. Equation (65) is the ordinary KL chain rule. Applying the same formulas to the m -mode and $(m-1)$ -mode marginals gives (66); Fisher information decreases under marginalization, so every bracket is nonnegative. $\square$

Proposition 4.3 (Hermite/Laguerre spectral refinement). *For the normalized OU equation (60), let $\mu = G_1 dw$ and $h_0 = q_0/G_1 - 1 \in L^2(\mu)$. Suppose that the Hermite coefficients of h_0 of total degree $1 \leq m < M$ vanish (the degree-zero coefficient vanishes by normalization). Then*

$$\chi^2(q_s \| G_1) := \int \left(\frac{q_s}{G_1} - 1 \right)^2 d\mu \leq e^{-Ms} \chi^2(q_0 \| G_1). \quad (67)$$

If the data are radial and mass, mean, and covariance are matched, the first unconstrained radial mode is Laguerre degree two (Hermite total degree four); thus a nonzero radial fourth cumulant is consistent with $M = 4$ and the e^{-4s} bound.

Proof. The OU semigroup multiplies a Hermite coefficient of total degree m by $e^{-ms/2}$. Parseval's identity therefore gives (67); in the radial sector, the Laguerre/Hermite correspondence and the moment constraints remove the lower sectors. $\square$

The OU result is a calibration, not a microscopic claim. It shows exactly how a conditional innovation can be resolved by mode order while the effective shape information decreases in time. In the radial benchmark below, the fourth cumulant decays as e^{-2s} , while Proposition 4.3 gives the sharper e^{-4s} decay for the corresponding χ^2 distance under its stated L^2 and moment hypotheses.

4.2 A common residual budget

Suppose a finite-resolution PDF obeys

$$\partial_t \rho_L = \frac{\sigma_L}{4} \Delta \rho_L - \frac{\Gamma_L}{2} \nabla \cdot (v \rho_L) + r_L, \quad \int r_L dv = 0, \quad (68)$$

with $\rho_L > 0$, $n_L > 0$, $\sigma_L \geq 0$. Let G_{n_L} be the matching Gaussian and

$$\varepsilon_L(t) := \int |r_L(t, v) \log \frac{\rho_L(t, v)}{G_{n_L(t)}(v)}| dv. \quad (69)$$

Proposition 4.4 (Duhamel stability). *If the displayed integrals, the second moment of r_L , and the Fisher information are finite, then, with $s_L(t) = \int_0^t \sigma_L(r)/n_L(r) dr$,*

$$H_L(t) + \int_0^t \frac{\sigma_L(r)}{4} I_{\text{rel}}(\rho_L(r) \| G_{n_L(r)}) dr \leq H_L(0) + \int_0^t \varepsilon_L(r) dr, \quad (70)$$

$$H_L(t) \leq e^{-s_L(t)} H_L(0) + \int_0^t e^{-[s_L(t)-s_L(r)]} \varepsilon_L(r) dr. \quad (71)$$

Proof. Writing $\mu_{2,L} = \int |v|^2 r_L dv$, one has $\dot{n}_L = \sigma_L + \Gamma_L n_L + \mu_{2,L}$. Differentiating the moving-reference entropy, including this variance contribution, gives

$$\dot{H}_L = -\frac{\sigma_L}{4} I_{\text{rel}}(\rho_L \| G_{n_L}) + R_L, \quad R_L = \int r_L \log \frac{\rho_L}{G_{n_L}} dv,$$

so $|R_L| \leq \varepsilon_L$. The logarithmic Sobolev inequality gives $\dot{H}_L \leq -\dot{s}_L H_L + \varepsilon_L$; multiplication by e^{s_L} and integration prove both inequalities. $\square$

Equations (47) and (71) have the same architecture:

$$\text{resolved information rate} = -\text{effective dissipation} + \text{residual}, \quad (72)$$

with the sign reversed when the resolved quantity is the hidden correlation reservoir rather than a shape defect. For hard spheres the residual is $R_{\text{contact}} + \alpha_N R_{\text{shift}}$; for a finite-box PDF it is the scalar budget ε_L . Neither residual can be discarded without a model-specific estimate.

5 A unified order–history–dissipation ledger

The two examples can be placed on a common two-dimensional information surface. At an observation level p , let Z_p be the resolved state, H_p the retained history, and Y_{p+1} the next layer. Define

$$M_p := I(Y_{p+1}; H_p \mid Z_p), \quad (73)$$

$$C_p := \mathbb{E} D_{\text{KL}}(P(dY_{p+1} \mid Z_p) \| Q_{\text{eff}}(dY_{p+1} \mid Z_p)), \quad (74)$$

$$D_p := \begin{cases} \frac{\sigma_p}{4} I_{\text{rel}}(\rho_p \| G_{n_p}), & \text{OU/PDF,} \\ \mathcal{D}_B[f_p], & \text{Boltzmann kinetic level,} \end{cases} \quad (75)$$

where Q_{eff} is the specified OU, WKE, or Boltzmann transition reference. The three entries have different units and are not to be added without a common time discretization. Their roles are:

- M_p measures dependence on the old history in the newly exposed layer;
- C_p measures closure error relative to the selected effective kernel;
- D_p is the nonnegative effective dissipation when the corresponding effective equation has one.

The memory and closure entries are not merely qualitative labels. If the actual conditional law is $P(dy | h, z)$ and its Z -conditional mixture is $P(dy | z)$, then

$$\begin{aligned} & \mathbb{E}_{P(h,z)} D_{\text{KL}}(P(dy | h, z) \| Q_{\text{eff}}(dy | z)) \\ &= I_P(Y; H | Z) + \mathbb{E}_{P(z)} D_{\text{KL}}(P(dy | z) \| Q_{\text{eff}}(dy | z)). \end{aligned} \quad (76)$$

This is the conditional KL chain rule in predictive form (with values in $[0, \infty]$). It separates a failure caused by retained history (M_p) from a failure of the Markov closure after history has been averaged out (C_p).

There is a useful common quadratic form behind the two dissipations. With $a = f f_1$, $b = f' f'_1$ and logarithmic mean (for $a, b > 0$)

$$\Lambda(a, b) := \begin{cases} (b - a)/(\log b - \log a), & a \neq b, \\ a, & a = b, \end{cases} \quad (77)$$

with the continuous extension $\Lambda(0, 0) = 0$ and the usual limiting interpretation on sets where one of a, b vanishes. The Boltzmann production can be written as

$$\mathcal{D}_B[f] = \frac{1}{4} \int B \Lambda(a, b) [\log b - \log a]^2 dx dv dv_1 d\omega. \quad (78)$$

This is a collision-graph analogue of the OU Fisher form $\sigma I_{\text{rel}}/4$: both are nonnegative costs of a log-density mismatch along the transitions retained by the effective generator. The analogy is structural, not an identification of the two generators.

For hard spheres, take the endpoint chain

$$Y_0 = \emptyset, \quad Y_s = (Z_1, \dots, Z_s) \ (s \geq 1), \quad Y_c^{(s)} = (Y_s, C_\delta(\mathcal{C}_{[0,t]})), \quad Y_h^{(s)} = (Y_c^{(s)}, H_{\text{coll}}^{(\delta)}), \quad (79)$$

where $(Y_s)_\# P = P_{N,s}$, $\mathcal{C}_{[0,t]}$ is the random collision record, and C_δ is a finite-resolution statistic of that record. The contact trace γ^- is a flux density (an intensity measure on contact times and marks), not by itself a normalized law of a random record; a record law additionally requires a sampling convention and normalization by the total flux. Each level literally contains the previous one. Theorem 3.4 concerns the bulk endpoint subchain. The contact and history levels have their own conditional KL increments; DHM's bulk estimate does not make them small. For a wave/PDF model, let $A = (A_k)_k$ be the mode-amplitude vector, Φ_{PDF} a specified measurable finite-resolution PDF block, Φ_{hi} a specified higher-order/resonant block, and $\mathcal{H}_{\text{res}}$ a finite-resolution resonant-history record. A compatible chain is

$$Y_0 = A, \quad Y_1 = (Y_0, \Phi_{\text{PDF}}), \quad Y_2 = (Y_1, \Phi_{\text{hi}}), \quad Y_3 = (Y_2, \mathcal{H}_{\text{res}}), \quad (80)$$

and Theorem 2.1 applies level by level. Replacing an observable by an unrelated moment would invalidate the monotonicity claim.

The ledger yields a practical criterion. A proposed kinetic arrow is credible only if (i) the closure budget C_p is shown to vanish or is bounded, (ii) the memory budget M_p is measured or controlled, and (iii) the effective dissipation D_p is compared with a collision- or mode-resolved estimator. Failure must be assigned to a specific mechanism (resonance degeneracy, hidden-variable dependence, finite-box recurrence, or closure failure), rather than to an undefined “loss of time reversal.”

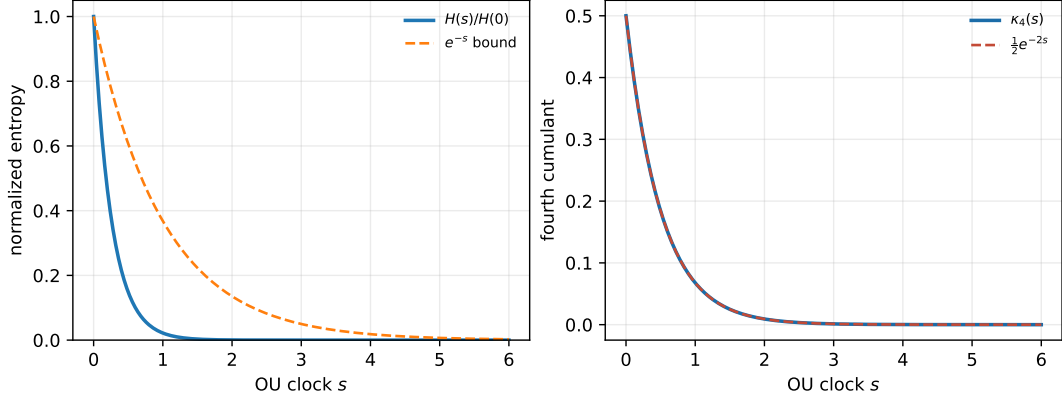


Figure 2: Analytic two-dimensional OU benchmark. The relative entropy to the unit-variance Gaussian lies below the rigorous $e^{-s}H(0)$ bound, while the radial fourth cumulant follows the exact $\frac{1}{2}e^{-2s}$ law. No finite-volume data enter this figure.

Table 2: Finite-density hard-disk audit quoted as an audit only. Values are means with independent-seed 95% intervals; m_2 is constrained by the preparation.

Observable	$N = 16, L = 8$	$N = 36, L = 12$
h_v	2.4907 ± 0.0279	2.6589 ± 0.0190
D_{KL} to Maxwell	0.3949 ± 0.0260	0.2140 ± 0.0261
$m_2 = \langle v ^2 \rangle$	2.0000	2.0000
$m_4 = \langle v ^4 \rangle$	6.9570 ± 0.3573	7.8817 ± 0.4573
$\kappa_4(v_x)$	-0.6522 ± 0.1742	-0.0270 ± 0.3155
$\langle v_i \cdot v_j \rangle_{r \leq 1.6}$	-0.1827 ± 0.0513	-0.0770 ± 0.0527

6 Analytic benchmark and finite-density audit

The OU benchmark used here is fully analytic. Take

$$q_0(w) = \frac{1}{2}G_{1/2}(w) + \frac{1}{2}G_{3/2}(w). \quad (81)$$

It has unit mass and unit radial second moment. With the radial convention $\kappa_4 = \mathbb{E}|W|^4 - 2(\mathbb{E}|W|^2)^2$, $\kappa_4(0) = 1/2$ and $\kappa_4(s) = \frac{1}{2}e^{-2s}$. Deterministic polar quadrature gives the entropy curve in figure 2; it is an illustration of Theorem 4.1, not evidence for a microscopic limit.

For completeness, the supplied hard-disk audit provides a reproducibility check for the microscopic code, not a Boltzmann–Grad convergence study. It used periodic boxes $(N, L) = (16, 8)$ and $(36, 12)$ at fixed area fraction $\phi = \pi/16$, four initial velocity beams, $T = 120$, and twelve independent trajectories. A 20×20 velocity histogram gave the following end-window summaries:

Short velocity reversal at $T = 2$ reproduced positions and velocities to below 2.8×10^{-13} , with relative energy error below 6×10^{-16} . Long reversal to $T = 120$ separated at box scale through floating-point amplification while conserving energy to about 10^{-14} . These observations are exactly what a reversible chaotic flow should show. They do not establish (37), (31), or a finite-density entropy law.

7 Falsifiable tests and required extensions

The synthesis makes measurements more specific than a generic entropy plot. A decisive Boltzmann–Grad study should:

1. scan $\varepsilon \downarrow 0$ and $N \uparrow \infty$ with $N\varepsilon^{d-1}$ fixed, while comparing to a Boltzmann solver or DSMC;

2. estimate the finite-resolution versions of $K_{N,s}$ and $\delta_{N,s}$ for several fixed s , and test $K_{N,s} \rightarrow 0$ alongside growth of $\mathcal{I}_N/N$;
3. sample incoming and outgoing collision velocities and compare the truncated surprisal sum

$$\widehat{\Sigma}_{N,T} = \frac{1}{N} \sum_{c:t_c \leq T} \log \frac{f(t_c, x_c, v_i^-) f(t_c, x_c, v_j^-)}{f(t_c, x_c, v_i^+) f(t_c, x_c, v_j^+)}, \quad (82)$$

with $\int_0^T \mathcal{D}_B[f_t] dt$;

4. measure the pair density both in the bulk and on contact, testing the trace norm in (48);
5. estimate bounded-observable variances and compare their collision part with the predicted Boltzmann–Langevin bracket [1]

$$\frac{d}{dt} \langle M(\varphi), M(\psi) \rangle_t = \frac{1}{2} \int B f f_1 \Delta \varphi \Delta \psi \, dx \, dv \, dv_1 \, d\omega, \quad \Delta \varphi = \varphi' + \varphi'_1 - \varphi - \varphi_1; \quad (83)$$

6. for wave systems, report the residual ε_L in (69), the conditional innovation B_m , and the effective clock s , rather than inferring an arrow from a single marginal histogram.

For a finite partition $\mathcal{P}_\delta$, first normalize the restricted probability vectors. In particular, the restriction $f_t^{\otimes s} \mathbf{1}_{\mathcal{D}_\varepsilon^s}$ has mass $1 - O_R(s^2 \varepsilon^d)$ on a fixed compact set by (16); the corresponding normalization must be included before applying Fannes–Audenaert (together with a positive reference-cell or bounded-log condition). The data-processing inequality then gives a lower-resolution bound, and the DHM L^1 estimate gives a rigorous fixed-resolution error bar. Refining $\delta \downarrow 0$ after taking the Boltzmann–Grad limit is a separate order-of-limits problem; this is precisely the experimentally accessible version of the hidden-arrow question. Related equilibrium and shorter-time fluctuation results provide context, but do not close the non-equilibrium long-time target posed here [1, 2, 11, 14].

8 Scope, limitations, and conclusion

The paper proves a structural statement, not a universal entropy law. The observation-chain theorem is an exact measure-theoretic identity. The hard-sphere order-escape result is conditional on the entropy/cross-entropy bridge and on a canonical transfer that is not contained in the bare factorial DHM estimate. The contact-trace estimate (48), a nonequilibrium Boltzmann–Langevin limit, and a coarse-grained endpoint large-deviation principle remain open. The OU and product-OU results are exact for their effective equations but require positivity, decay, finite Fisher information, and (for spectral refinements) an L^2 density ratio. The finite-density audit has fixed packing fraction and cannot be used as evidence of the Boltzmann–Grad limit.

Within these boundaries, the synthesis resolves the apparent paradox in a testable way. Three operations do not commute: evolve the full reversible law, discard collision- or resonance-connected information, and take the kinetic limit. Along the observation axis, each refinement reveals a nonnegative conditional KL. Along the time axis, a closed hard-sphere system moves one-particle information into an order-summed correlation reservoir, while an effective OU closure dissipates shape information at a Fisher rate. The macroscopic arrow is the part of this two-dimensional ledger that survives the chosen projection and reference dynamics. Its location can be measured, and the residual required to falsify it is explicit.

Authorship and provenance. This is a synthesis draft prepared from the two supplied manuscripts, one on collision-history information in the Boltzmann–Grad limit and one on order-resolved entropy in kinetic limits. It is not a paper by Deng, Hani, or Ma, and no priority is claimed. Actual contributors must verify every proof, bibliography entry, numerical datum, and authorship decision before any submission.

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